Supplementary File

Integration of photovoltaic modules with phase change materials: experimental testing, model validation and optimization

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In this supplementary material, the procedure for the evaluation of the photovoltaic (PV) degradation factors, the phase change material (PCM) thermal characterization, and the sensor's technical characteristics are described, as well as the results of the comparison between the modeled and measured temperatures in the case of the reference PV module and PV-PCM29 module are illustrated. Additional tables and figures that are used for the development of the experimental setup to test a traditional PV module and an enhanced PV module integrated with PCMs are provided. The explanations of how the procedures, the tables, and the figures in this Supplementary File are used for the study are illustrated in the main paper.

S1. Procedure for the evaluation of the photovoltaic degradation factors

In order to evaluate the degradation of each PV module, a specific preliminary test campaign was designed to assess the actual reference parameters in STC. The Current-Voltage (I-V) and the Power-Voltage (P-V) curves of each PV module were measured outdoors under clear sky conditions, within an irradiance range between 600 W/m² and 1000 W/m²; 7 I-V were measured for each module by using the I-V tracer I-V500w manufactured by HT [S1]. The I-V500w is an instrument intended for testing PV modules and PV strings up to 1500V/10A or 1000V/15A, allowing the evaluation of whether the PV module or the PV string fulfills the parameters declared by the manufacturer. Once the I-V curve in Operating Conditions (OPC) is measured, together with the plane of array irradiance and the temperature on the back of the PV module, the I-V curve translated to the Standard Test Conditions (STC) is automatically calculated by the instrument and the compliance with the specifications declared by the module manufacturer was checked. The measurements and the results provided by the I-V tracer are compliant with the IEC/EN60891 (I-V curve test) [S2] and IEC/EN 60904-5 (temperature measurement) [S3], but the procedure for correcting measured I-V curves to other conditions of temperature and irradiance is not declared by the manufacturer and the input parameters, such as series resistance and temperature coefficients, are related to a PV module in perfect factory condition.

To validate the actual PV module parameters in STC provided by the I-V tracer, 3 additional methods have been implemented for the translation of the I-V curves, or their relevant points, measured in OPC to STC.

Method 1 is based on the usual equation for the translation of the open circuit voltage, the short circuit current, and the maximum power, as reported in Eqs. (S1)-(S3):

$$I_{sc,OPC} = \frac{G_{OPC}}{G_{STC}} I_{sc,STC}\left[1 + \alpha_{Isc}\left(T_{c,OPC} - T_{c,STC}\right)\right] \tag{S1}$$

$$V_{oc,OPC} = V_{oc,STC}\left[1 + \beta_{Voc}\left(T_{c,OPC} - T_{c,STC}\right)\right] + A\, ln\left(\frac{G_{OPC}}{G_{STC}}\right) \tag{S2}$$

$$P_{mpp,T} = \frac{P_{mpp,OPC}}{\left[1 + \gamma_{Pmpp}\left(T_{c,OPC} - T_{c,STC}\right)\right]} \tag{S3}$$

where T_c is the cell temperature, G is the tilted irradiance, N_s is the number of cells in series within a PV module, k is the Boltzmann constant, q is the elementary charge, and the subscripts OPC and STC identify the operating and standard conditions. This approach applies to only three specific values and requires the knowledge of the temperature coefficients, which are known only for the PV module in perfect factory condition. Consequently, the results are valid only assuming that the temperature coefficients are not affected by aging processes.

Method 2 is based on the PV module one-diode equivalent circuit, also known as the five-parameter model. Nonlinear least square fitting was applied to extract the model parameters from the I-V curves measured in OPC. Then, model parameters were adjusted to represent the operation in STC [S4]. Finally, the PV module I-V curve in STC was calculated and the reference parameters representing the PV actual performance were derived.

Method 3 applies the standard IEC/EN60891 [S2] for the correction of the I-V curves measured in OPC. In particular, Correction Procedure 4 reported by the previous standard is the only one that can be applied to the available dataset, and it is used to calculate the I-V curve in STC. As for Method 2, the reference parameters representing the PV's actual performance were derived from the PV module I-V curve referred to as STC.

For each PV module and each correction procedure, the electrical data extracted from each I-V curve in STC were averaged. Since there are no specific reasons to prefer a particular correction procedure, the actual electrical data in the STC of each PV module were defined as the average of the electrical data in STC obtained for the single procedure. The degradation factors (DFs) were defined for the whole set of electrical data in STC as the difference between the actual value and the same value in perfect factory condition, as a percentage of the latter. Table 2 in the main manuscript shows, for each PV module involved in the test campaign, the electrical data in STC derived from each correction procedure, the actual electrical data in STC, and the DFs.

S2. Phase change material thermal characterization

Building upon prior research [S5] and utilizing a Differential Scanning Calorimeter (DSC), various parameters of the InfiniteR29, such as melting and freezing points, heat capacity, and latent heat, were comprehensively determined across the entire thermal cycle. As shown in the DSC curve reported in Figure S1, the analysis revealed a 4.3 °C disparity between the peak temperatures of melting and freezing. Specifically, the melting point was identified as 33.47 °C, while the freezing point was recorded at 29.1 °C.

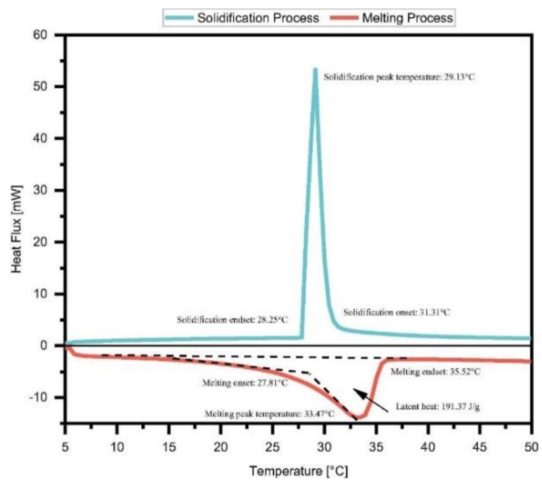

Figure S1. DSC results for InfiniteR29 [S5].

The phase transition during the melting process spanned approximately 8 °C, whereas during freezing, it slightly exceeded 3 °C. The energy absorbed during the melting process was quantified at around 191.37 kJ/kg. Additionally, the specific sensible heat capacity, calculated as the change in enthalpy before the phase change, was established as 3.17 kJ/(kg K) for the melting process. In relation to thermal conductivity, the liquid sample manifested a value of 0.64 W/(m K), whereas the solid sample exhibited a value of 0.83 W/(m K) [S5]. The density of the PCM was further evaluated at temperatures of 20 °C, 30 °C, and 45 °C, employing a pycnometer with a volume of 24.985 cm³.

Since the three PCMs are identical in their chemical nature, the DSC experimental curves available in the literature for PCM29 were used to reconstruct the curves for the other two PCMs (see Figure S2) accounting for the hysteresis phenomenon. Specifically, nonlinear regression was applied to derive the function linking the heat flux to temperature in both the melting and solidification phases according to Eqs. (S4) and (S5):

$$\varphi_{sol}(T_{PCM}) = \sum_{i=1}^{5} a_i \cdot exp\left[-\left(\frac{T_{PCM} - b_i}{c_i}\right)^2\right] \tag{S4}$$

$$\varphi_{mel}(T_{PCM}) = \sum_{i=1}^{5} d_i \cdot exp\left[-\left(\frac{T_{PCM} - e_i}{f_i}\right)^2\right] \tag{S5}$$

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The obtained coefficients are reported in Table S1. The specific heat capacity as a function of temperature was determined by considering the PCM mass starting from the DSC curves.

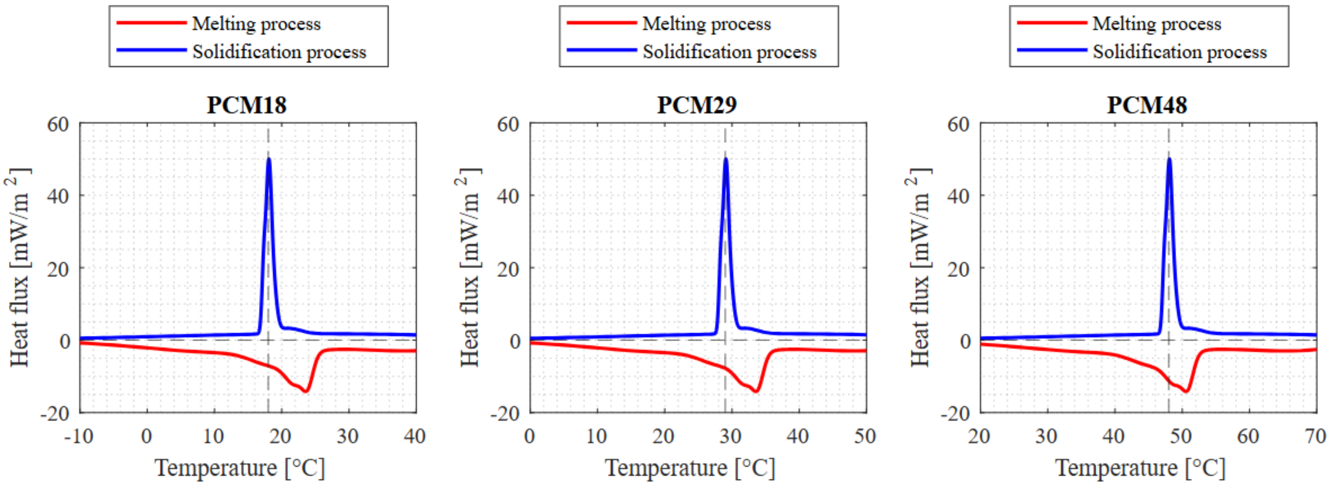

Figure S2. PCM melting and solidification curves.

Table S1. Coefficients of the non-linear regression curves of the DSC tests for the three PCMs.

	Solidification function coefficients				Melting function coefficients		
	PCM18	**PCM29**	**PCM48**		**PCM18**	**PCM29**	**PCM48**
a_1	36.77	36.77	36.77	d_1	-3.51	-3.51	-3.51
b_1	18.03	29.03	48.03	e_1	13.08	23.08	40.08
c_1	0.5488	0.5488	0.5488	f_1	18.65	18.65	18.65
a_2	16.01	16.01	16.01	d_2	-3.834	-3.834	-3.834
b_2	17.32	28.32	47.32	e_2	18.88	28.88	45.88
c_2	0.3781	0.3781	0.3781	f_2	4.309	4.309	4.309
a_3	16.31	16.31	16.31	d_3	-7.055	-7.055	-7.055
b_3	18.57	29.57	48.57	e_3	22.16	32.16	49.16
c_3	0.7888	0.7888	0.7888	f_3	2.091	2.091	2.091
a_4	1.547	1.547	1.547	d_4	-6.348	-6.348	-6.348
b_4	20.92	31.92	50.92	e_4	23.91	33.91	50.91
c_4	2.596	2.596	2.596	f_4	1.081	1.081	1.081
a_5	1.761	1.761	1.761	d_5	-2.499	-2.499	-2.499
b_5	25.47	36.47	55.47	e_5	40.15	50.15	67.15
c_5	32.28	32.28	32.28	f_5	10.62	10.62	10.62

S3. Sensor's technical characteristics

Weather station

Figure S3. Weather station and measurement system at SolarTech^{LAB} [S6].

Table S2. Solar irradiance and temperature sensor characteristics [S6].

Irradiance sensors (Pyranometer (LSI, DPA252))	
Standard	Secondary standard ISO 9060
Measurements range (W/m^2)	< 2000
Spectral range	300-3000 nm
Total achievable daily uncertainty	< 2 %
Directional response	<±5.4 W/m^2
Thermal drift	< 2 %
Temperature and humidity sensor (LSI, DMA 875)	
Temperature sensor	Pt100 1/3 B (DIN EN 60751)
Measurements range	[-30°C, +70°C]
Uncertainty	0.2 °C (at 0 °C)
Resolution	0.04 °C
Response time (T90)	3 min: with filter; 20 s: without filter (air speed 0.2 m/s)

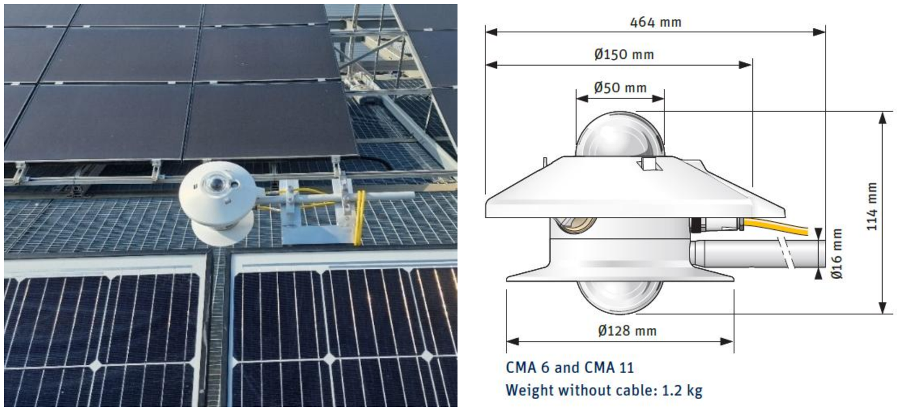

Figure S4. Albedometer installed on the experimental setup [S7].

Data logger for temperature and heat flux

Table S3. Data logger specifications [S8].

Manufacturer	HIOKI E.E. Corporation
Description	Value by the Manufacturer
Analogue inputs	16 differential 24-bit
Input ranges	± 100 V DC
Accuracy	± 10 μV with -10 mV ÷ 10 mV range ± 50 mV with -100 V ÷ 100 V range
Inputs type	Voltage and temperature
Recording interval	10 ms ÷ 1 h

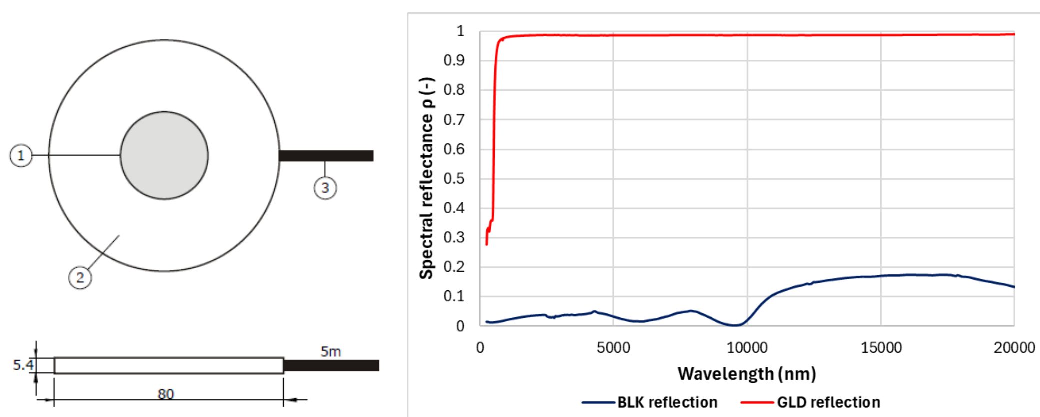

Figure S5. Heat Flux Sensor: (1) sensing area, (2) passive guard of ceramics-plastic composite, (3) cable; spectral reflectance of HFP01 sensors with BLK-80 black sticker and GLD-80 gold sticker* [S9].

* *The BLK black sticker has an average reflectance of approximately 3 % (ρ_{BLK} = 0.03) across the ultraviolet (UV), visible, near-infrared, and far-infrared spectra. Conversely, the GLD gold sticker exhibits a reflectance of about 35 % (ρ_{GLD} = 0.35) in the UV range, which increases progressively across the visible*

spectrum, reaching an average reflectance of 98 % (ρ_{GLD} = 0.98) in the near-infrared and far-infrared regions.

Table S4. Heat flux sensor specifications [S9].

Manufacturer		Hukseflux
Measurand		heat flux
Description	Symbols/Units	Value by the Manufacturer
Sensing area	A_{HFM} [m^2]	8 x 10^{-4}
Guard width-to-thickness ratio	[m/m]	5
Sensor thermal resistance	[K/(W/m^2)]	71 x 10^{-4}
Sensor resistance range	[Ω]	1 to 4
Sensor thickness	[m]	5.4 x 10^{-3}
Uncertainty of calibration	[%]	± 3 (k = 2)
Measurement range	[W/m^2]	-2000 to 2000
Sensitivity (nominal)	[V/(W/m^2)]	60 x 10^{-6}
Rated operating temperature	[°C]	-30 to +70
Cable diameter	[m]	4 x 10^{-3}
IP protection class		IP67
Standard cable length	[m]	5

Table S5. Thermocouple specifications [S10].

Manufacturer		Tersid
Measurand		Surface temperature
Description	Symbols/Units	Value by the Manufacturer
Type		T
Model		TEX/TEX-30-TT
Mode		Differential
Measurement Range	[°C]	(-220) ÷ (+400)
Sensitivity	[µV/◦C]	48.2
Calibration Curve		
Conductors		Monotrefol
Insulation		Teflon FEP (- 200/ + 200°C)
Parallel pair conductor diameter (30 AWG)	[mm]	0.25
External dimensions	[mm]	1.2 x 1.8
Calibration curve		[S11]

S4. Additional figures for the validation of the PV-PCM model: temperature comparison

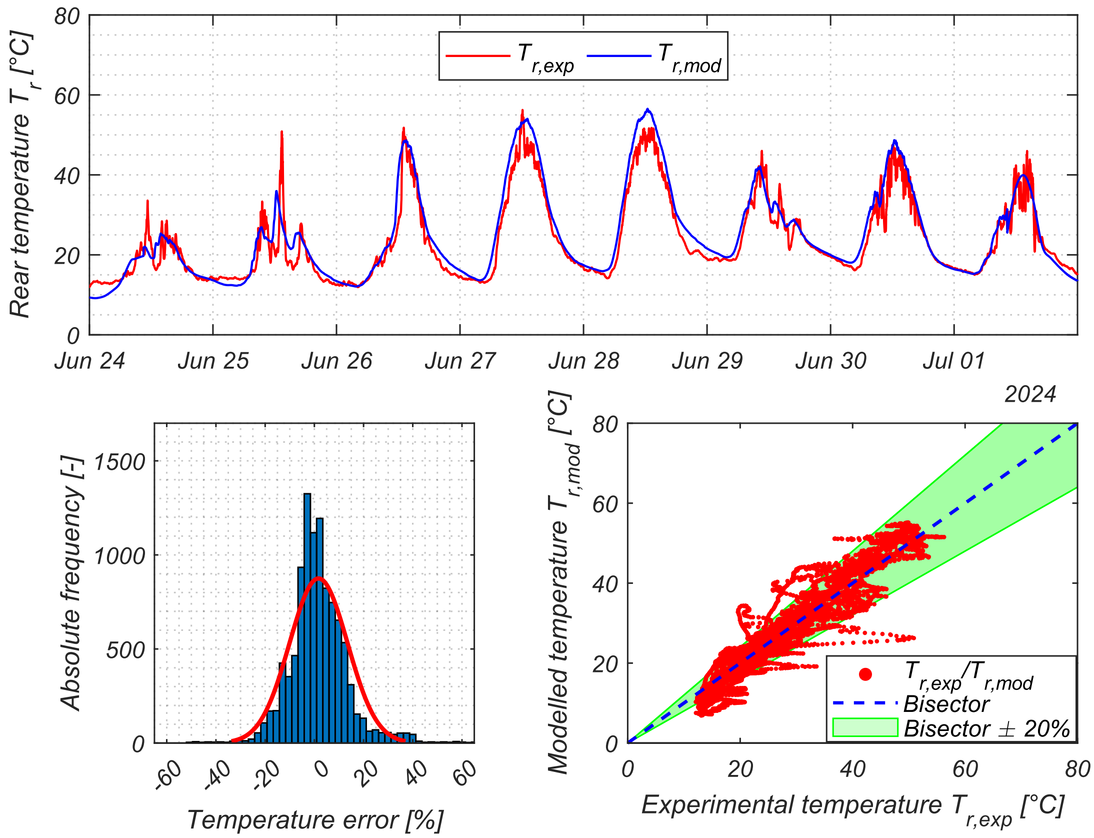

Figure S6. Comparison of the trend of the modeled and experimental rear temperature, rear temperature error distribution, and scatter plot of experimental and modeled temperature. Reference PV module.

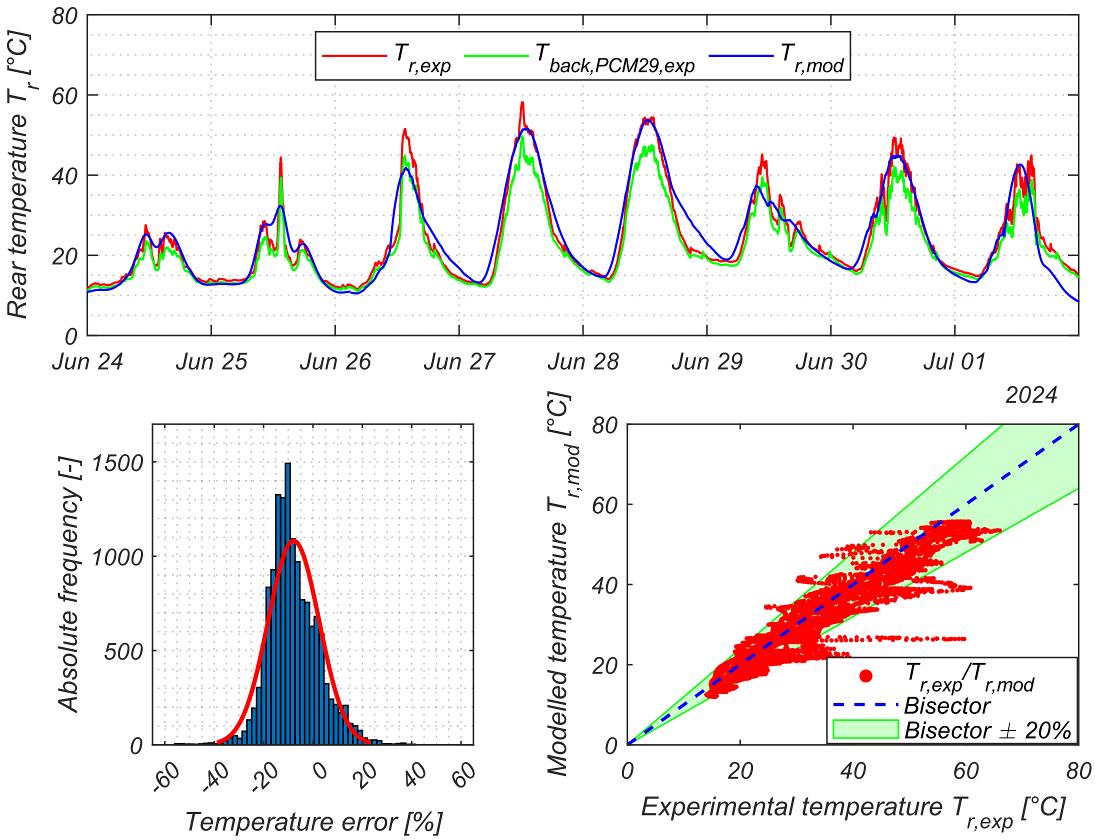

Figure S7. Comparison of the trend of the modeled and experimental rear temperature, rear temperature error distribution, and scatter plot of experimental and modeled temperature. PV-PCM29 module.

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